Setup for FDM 320

Welcome to the Tooling University. This course is designed to be used in conjunction with the online version of this class. The online version can be found at https://www.toolingu.com. We offer high quality web-based e-learning that focuses on today's industrial manufacturing training needs. We deliver superior training content over the Internet using text, photos, video, audio, and illustrations. Our courses contain "roll-up-your-sleeves" content that offers real-world solutions on subjects such as Metal Cutting, Workholding, Materials, and CNC with much more to follow. Today's businesses face the challenge of maintaining a trained workforce. Companies must locate apprenticeship programs, cover travel and lodging expenses, and disrupt operations to cover training needs. Our web-based training offers low-cost, all-access courses and services to maximize your training initiatives.

© Copyright 2023 Tooling University, LLC. All rights reserved.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Class Outline

Fused Deposition Modeling
Networking and Software
FDM Machine Configurations
Key Machine Components
Machine Calibration
Software Configuration
Build Surface Setup
FDM System Review
Build Chamber Setup
Sensors and Alarms
Part Removal
Setup Approaches Review

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Lesson: 1/12

Fused Deposition Modeling
The FDM process is used in small desktop hobbyist, or consumer-level, machines as well as industrial-level machines with a large build area. Desktop FDM machines typically work better for low volume tasks such as rapid prototyping, which enables an engineer to quickly create several prototypes in a short time period. In contrast, industrial FDM machines generally work better for end-use parts or products, which rely on repeatable, consistent results for each production run.

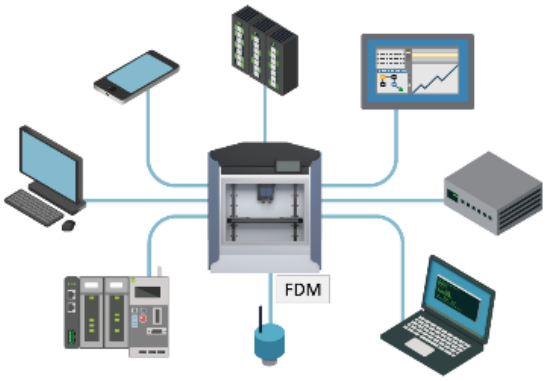
An important process in industrial FDM printing is setup, which focuses on preparation, changeover, and adjustment of the FDM machine during production. Preparation for FDM can include gathering and organizing materials to enable an efficient printing. Changeover typically involves switching materials, which can require cleaning and preheating that may lead to longer downtimes in production if done incorrectly. Adjustment in FDM printing might involve changing the temperature of the extrusion head or the speed of the print to ensure part accuracy. Although often a time-consuming process, proper setup can prevent costly errors and increase production efficiency.

This material is based upon research supported by, or in part by, the U.S. Office of Naval Research under award number N00014-19-1-2742.

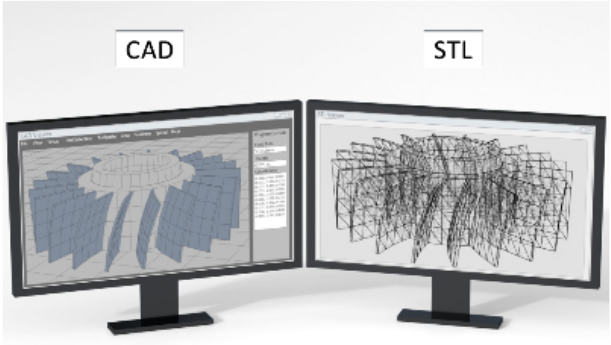
Video content available online.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

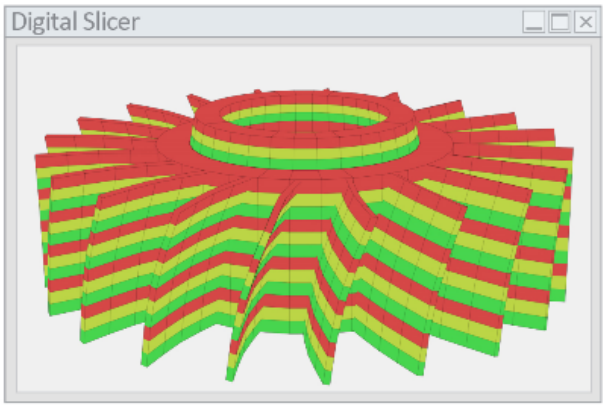
Networking and Software



FDM machines that connect to the IIoT can directly send and receive data without human intervention, which is known as machine-to-machine (M2M) communication. Monitoring the machines through the IIoT can help monitor prints and detect errors, which an operator can catch in real time during part production. This can help minimize waste associated with cost and time.



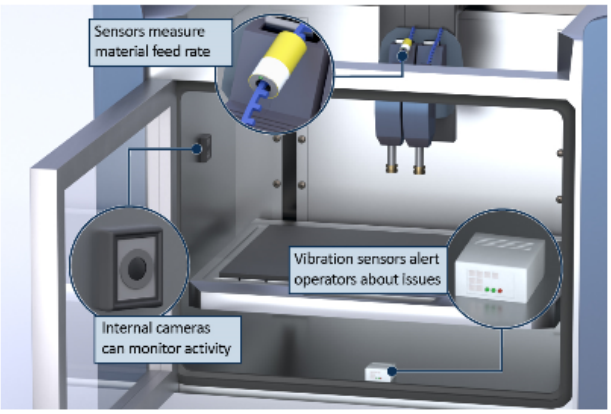
Like other AM methods, engineers using FDM machines work with software, often computer-aided design (CAD), to create a 3D design file of a part. The CAD design file is a digital representation of the physical part, but FDM machines cannot directly read design files. Because of this, an engineer or designer must convert the 3D design file into an STL format file, one of the most widely used file formats for 3D printing. The STL format uses a sequence of triangles to approximate the 3D model's external surface.



Digital slicer software takes the STL file information and "slices" it, or translates it into the layers. Many FDM machines will already have the slicer software installed. Slicer software takes the layer information and creates G Code, a numerical programming language that controls the

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

functions and motions of the FDM hardware. When setting up a job for an industrial FDM machine, it may build many different parts in the same build. However, the material, layer thickness, and nozzle parameters need to be the same for all jobs.

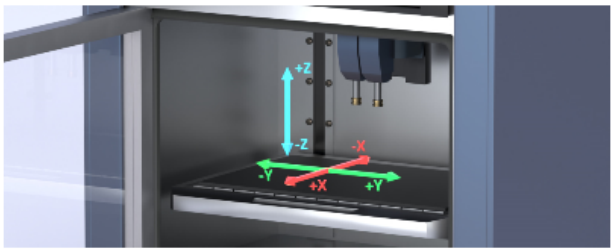


M2M communication on the IIoT utilizes sensors that collect and analyze data from FDM machines and transmit it to other equipment or devices. For example, sensors can pick up unusual vibrations in the printer head or measure the speed of the material feed. The IIoT utilizes this real-time data to alert an operator to fix the vibration issue. These and other smart manufacturing processes generate a large amount of data.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Lesson: 3/12

FDM Machine Configurations



The configuration of an FDM machine refers to the location and movements of its components. With some exceptions, most industrial size FDM printers use the Cartesian coordinate system to define and program the locations of machine components and part features. On these machines, configuration describes how the machine components move within the Cartesian coordinate system. FDM machines using the Cartesian coordinate system may have different mechanical designs. For example, in some FDM systems the nozzle is placed at the center of three arms. However, FDM systems are united in the way they all use three linear axes, labeled X, Y, and Z, to define the location of the components in physical space. The nozzle and machine table move in one or more of the axes to deposit the build materials using a rectangular grid pattern.

FDM machine manufacturers typically configure machines by programming the X,Y, and Z axes into them. The axes may align with different components on different FDM machines, depending on how the machine moves. Most FDM machines use an XY-head Z-bed configuration. This means that the machine bed moves only in the Z axis, or up-and-down, while the deposition head moves back-and-forth and left-to-right in the X and Y axes. On other machines, the components may move in different axes, or use a system other than Cartesian coordinates. For example, some printers use polar, or circular coordinates.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Key Machine Components
The basic setup operations for an industrial FDM machine revolve around three key elements: The build chamber, material extruder, and the build surface.

The build chamber of an FDM machine, which is also referred to as an oven, locks to control the temperature of the part build. The type of materials used in a build determine the correct temperature of the build chamber. Opening the chamber door during an FDM operation or cool-down will alter the temperature and may affect the part build. To avoid part defects, FDM machines lock the chamber until the production cycle completes.

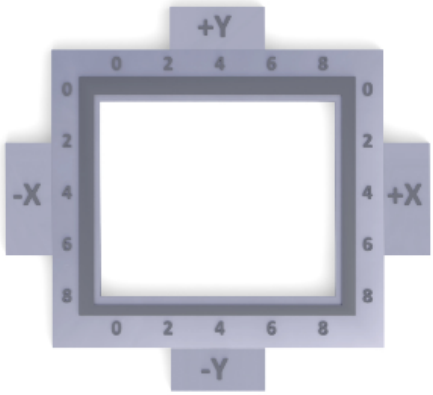
The extruder melts and dispenses build and support material. The extruder assembly includes material feedlines, which feed the raw material to the extruder head, material canisters, which contain the raw build or support material, and nozzle tips, which precisely direct and dispense material during the build. Without regular cleaning during setup and maintenance, the material feedlines, canisters, and nozzle tips may clog during a print.

The build surface is the flat physical surface upon which the FDM machine builds the part. Build surfaces for industrial FDM typically utilize either a build sheet or build tray, which sit above the flat, built-in surface of the machine called the platen. A debris chute or purge chute often collects excess waste from the build surface and the extruder areas. Depending on the machine type, waste collected from the purge chute is generally deposited in the purge bucket or the bottom of the build chamber.

Video content available online.

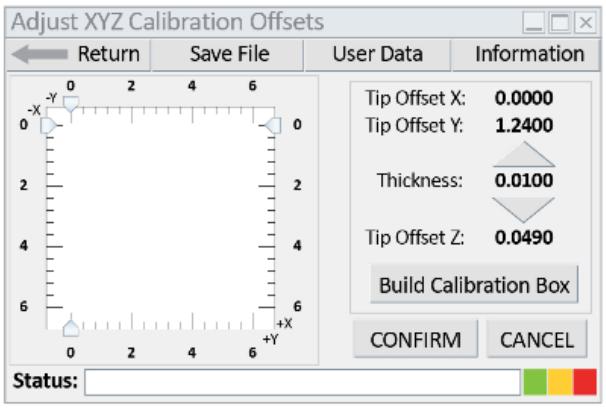
Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Machine Calibration



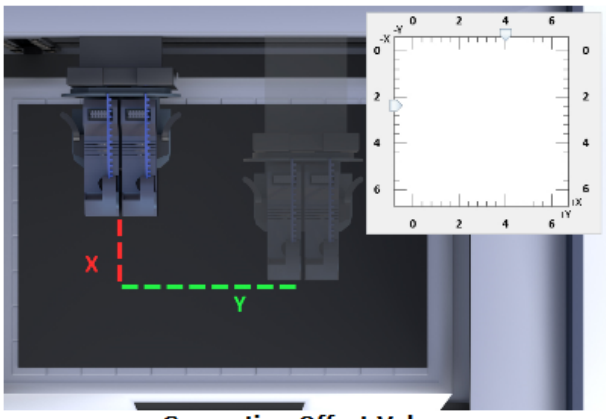
Example of a Calibration Print

The extruder of an FDM machine has at least one nozzle that deposits build material, referred to as the model head, and at least one nozzle that deposits support material, referred to as the support head. Running a test print, or calibration print, enables an operator to manually inspect the ability of the machine to print to design specifications. Manually inspecting the test part for errors will help to detect any issues with the alignment of the model head and support head.



Calibration Automation

For three axis FDM machines, the calibration process is mostly automated. To confirm calibration, an operator should run a test print, or calibration job, of the part near the center of the platen before performing the full part run. Afterwards, an operator will need to manually inspect the test part using a magnification device.

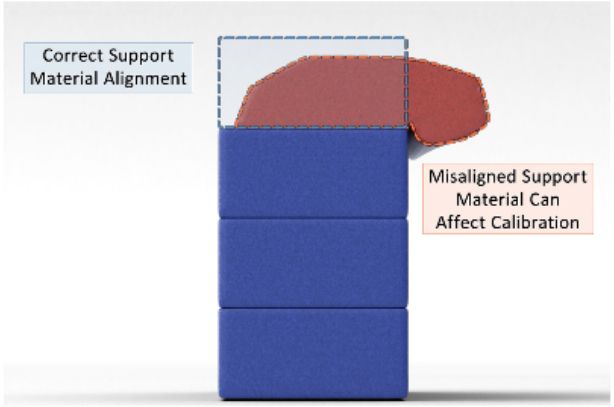


Computing Offset Values

To inspect the calibration job, first verify that the machine has correctly aligned the support material with the build material. Then, verify that the machine deposited the materials in the right locations along the X and Y axes. If not, it may be necessary to adjust the offset value for

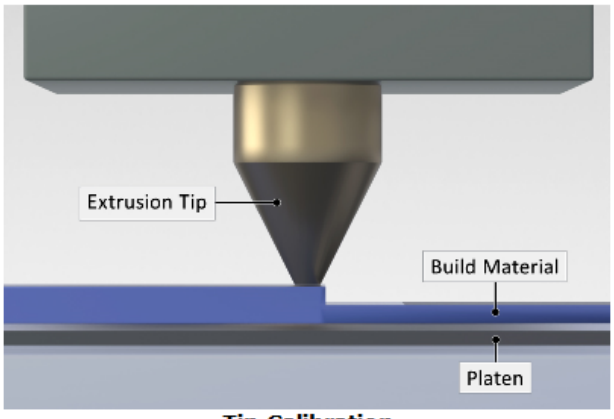
Copyright © 2023 Tooling U, LLC. All Rights Reserved.

the print head or build surface. Offsets are numerical values stored in the FDM control that reposition machine components to the correct location for a specific part run. Some FDM machines will print the offset values directly on the calibration print.



Support Material Calibration

After determining that the test part is within parameters in the X and Y axes, measure and verify the calibration part's thickness, or Z axis, using a caliper or micrometer. Additionally, it is important to also calibrate support material. If the machine misaligns the support material during calibration, it could cause it to sag and affect the part's support material measurement.



Tip Calibration

Other factors that may affect calibration can include the material type and the distance of the extrusion tip from the platen. For example, if maintenance personnel must exchange material tips, proper calibration will include updating the tip offset. This ensures that the new tip will begin at the precise height.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Software Configuration

After calibrating the machine, an engineer or operator must arrange the software, or configure the software settings, to ensure that the job file is compatible with the machine's digital slicing software system settings. For example, it is important to confirm that the material loaded in the machine matches the material type designated in the job file. During the configuration process, an operator can also verify other basic information, such as the material amount needed, layer heights, and the tip size of the extruder. An operator may also need to delete previous jobs in the queue on the machine's hard drive to avoid accidentally printing the wrong file.

After configuring the settings, an operator will typically need to confirm the external and internal parameters of the part build. Internal parameters include the interior of the part, which may or may not include a fill pattern. For example, by using lightweighting techniques, an engineer can design a part with a lattice structure interior that retains the same strength as a solid interior, but uses less material and prints faster. Visible parameters include build layer thickness, or layer height, of the part. For instance, selecting a larger layer height may increase print time without sacrificing part quality.

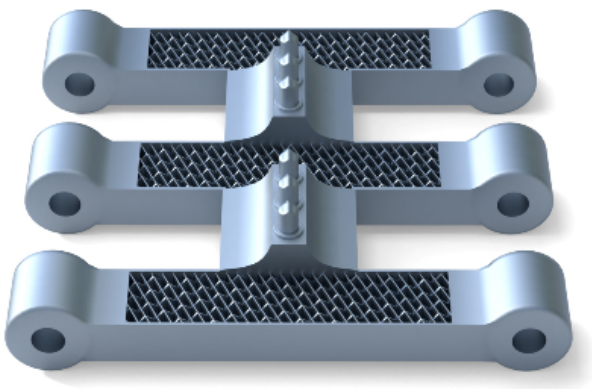


Figure 1: Lattice Structure

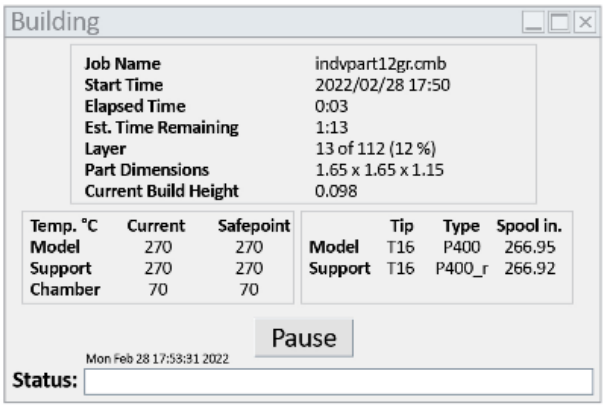
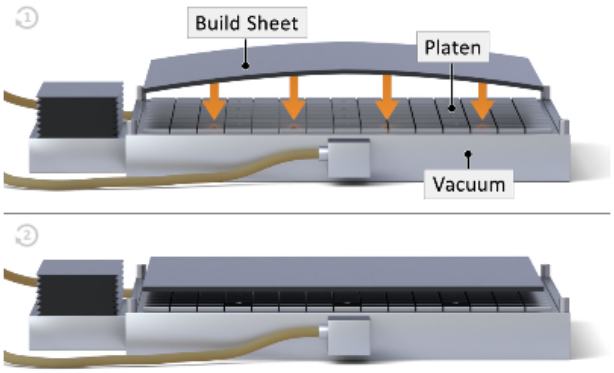


Figure 2: Example of a Job File

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Build Surface Setup



Some FDM machine build surfaces contain a vacuum, which holds items in place using air pressure. These build surfaces utilize a platen. A platen's surface contains holes or grooves that enable the vacuum's lower air pressure to pass through it, which allows it to securely hold a build sheet. A build sheet is a thin piece of material that rests on the platen in the build chamber. Other FDM machines contain a solid build tray, rather than a vacuum, as the build surface. Machines that have a build tray surface might utilize a disposable build plate, which is similar to a build sheet but is thicker and more rigid.

The purpose of the build sheet or plate is to provide a flat surface that a part can temporarily adhere to during the part build, rather than adhering to the build surface itself. Build sheets and plates can also act as a buffer layer to help prevent any dirt or debris within the machine from affecting the final part dimensions. Additionally, they can help prevent wear and tear to the build surface during part removal.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

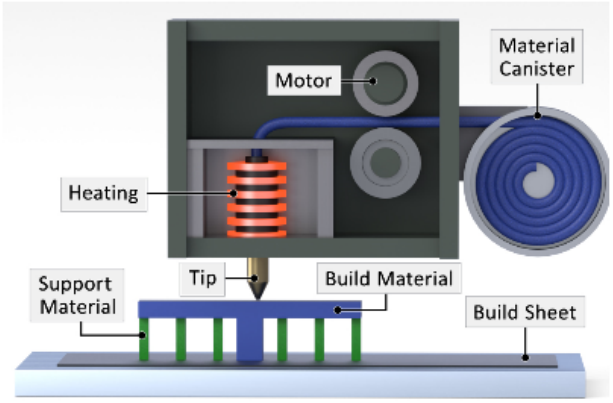
Lesson: 8/12

FDM System Review

1. FDM machines can typically read design files directly. **FALSE**
2. Desktop FDM machines generally work better for end-use parts or products, which rely on repeatable, consistent results for each production run. **FALSE**
3. External parameters include the layer thickness. **TRUE**
4. FDM machines increasingly use digital networks and software to connect to the IIoT. **TRUE**
5. A platen typically holds a build sheet on an FDM machine. **TRUE**

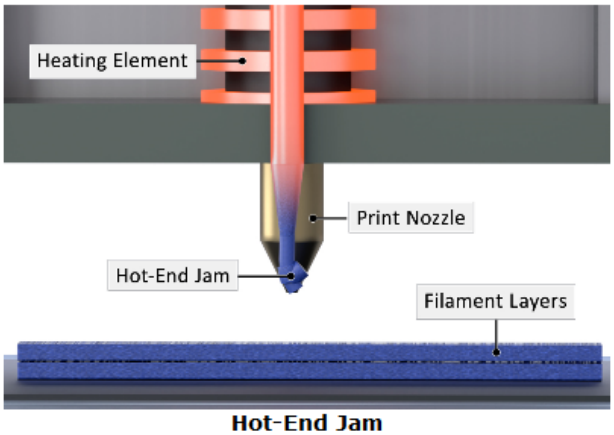
Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Build Chamber Setup



Typical Build Chamber

First, examine the material canisters to make sure that they are loaded correctly and contain the material specified for the job build. Although it may vary by machine, typically the machine will create an alarm on the display screen if the material does not match the job build instructions. However, operators should also check the job sheet and part specifications to confirm the material. In addition, verify that the cold parts of the support material and part material are loaded tightly to the hot part of the extruder. Any air gaps may cause a defect in the part build.



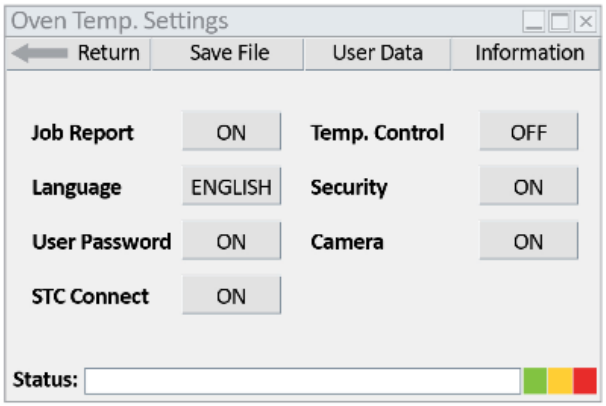
Hot-End Jam

Next, examine the extruder nozzles, which are a critical part of an FDM printer's ability to create an accurate part. Nozzles can clog, which causes material to extrude incorrectly and create inaccurate part dimensions. Even a slightly dirty nozzle can affect the quality of a print, so they must remain free of any debris or material. For example, even if the lower part of the nozzle appears clean, a hot-end jam may occur near the upper part of the nozzle. In addition, a nozzle may start off clean but accidentally pick up material while depositing a layer, which may affect part quality.

Purge Bucket

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Additionally, check the purge bucket before each job build and empty it if needed. The purge bucket, typically located under the automatic cleaning assembly in the heated build chamber, collects waste and debris from print jobs and routine maintenance.

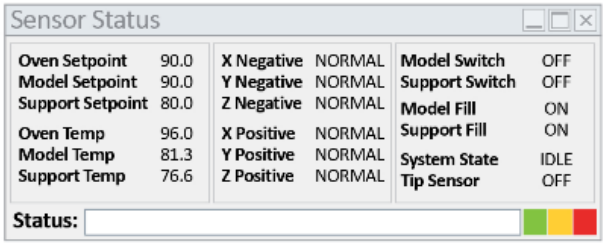


Temperature Display

Finally, before starting or resuming any building operation, verify that the build chamber has reached optimal temperature. On some FDM machines, it may take up to four hours to reach optimal building temperature.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Sensors and Alarms



For example, sensors can confirm the oven and material temperatures, vacuum status, motor status, and extruder tip conditions, among other variables that may occur during a job build. Maintaining accurate sensors helps ensure repeatability during FDM part builds for end-use production.

When a sensor detects an issue with the FDM machine's functionality, it will create an alarm to alert operators and maintenance personnel. A common alarm that may occur during the beginning of a print may involve incorrect vacuum pressure, which can result from a build sheet placed improperly on the platen or debris in the vacuum line. Another problem can involve improper material loading. Before restarting the print, make sure the proper amount of time has passed for the extruder to heat the material.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Part Removal

After completing the job build, an operator will need to open the door of the FDM machine and remove the build tray or sheet from the heated chamber with the part still attached. Removing the part while it is still in the build chamber may damage the platen. Some FDM machines can reuse build sheets or trays for some job builds. However, it is generally advisable to install a new build sheet or tray with each job build. For example, a reused build sheet may contain material residue from the previous print, which may affect the new print or damage the platen if turned over.

For some materials, an FDM machine may perform an automatic or extended cooling period after the job build. However, some build materials, such as Polyphenylsulfone (PPSF), detach more easily while the build tray or sheet is still hot. For these materials, an operator should remove the build sheet from the heated chamber as soon as possible. If a part does not detach easily, the operator may need to use a scraper device to carefully separate the part from the build sheet. When removing the build sheet, an operator should take care to wear the proper personal protective equipment (PPE), which should include leather gloves and goggles.

Figure 1: Personal Protective Equipment (PPE)

Figure 2: Build Sheet

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Lesson: 12/12

Setup Approaches Review

- **Build Tray**: Provides a flat surface that a part can temporarily adhere to during the part build.
- **Purge bucket**: Collects waste and debris.
- **Sensors**: Can confirm the oven and material temperatures.
- **Scraper**: Device to carefully separate the part from the build sheet.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

Class Vocabulary

3D Printing	The process of producing a 3D object using a specialized tool that creates successive layers of material. 3D printing technically must involve the use of a print head or nozzle, but the term is often used interchangeably with additive manufacturing.
Additive Manufacturing	AM. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.
AM	additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. AM methods typically build up layers of material to create an object.
Axes	An imaginary line or circle that is used to define the position of an object in space. The linear axes are X, Y, and z, and the rotational axes are often designated using the letters A, B, and C.
Build Area	The location and space within an additive manufacturing machine in which the part is actually constructed. Build areas in some additive manufacturing machines are open while others are surrounded by an enclosure.
Build Chamber	The area contained within an additive manufacturing (AM) machine where the part is actually built. Many build chambers are enclosed and heated to improve the safety and quality of the manufacturing operation.
Build Sheet	A thin piece of material that rests on the platen in the build chamber. Build sheets provide a flat surface that a part can temporarily adhere to during the part build, rather than adhering to the platen itself.
Build Tray	A thick sheet of material that rests on the platen in the build chamber. Build trays, which are typically sturdier than build sheets, provide a flat surface that a part can temporarily adhere to during the part build, rather than adhering to the platen itself.
Calibration	The comparison and adjustment of a device with unknown accuracy to a device with a known, accurate standard. Calibration eliminates any variation in the device being checked.
Caliper	A measuring instrument with a pair of jaws on one end and a long beam containing a marked scale of unit divisions. Calipers can measure both internal and external features.
Cartesian Coordinate	A numerical system that describes the location of an object by expressing its distance from a fixed position along three linear axes. Cartesian coordinates are used to direct machine tool movements.
Computer-Aided Design	CAD. Computer software used to create a 3D model of a part before it goes into prototyping or production. Computer-aided design models are converted to an STL format for use by additive manufacturing machines.
Data	A collection of numbers, facts, and information about a process or product. Data can be created, communicated, and recorded by sensors in smart objects.
End-Use	Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.
Extruder	A nozzle or mechanism that shapes and dispenses semi-solid material. Extruders are used in fused deposition modeling (FDM).
FDM	Fused deposition modeling. An additive manufacturing process that builds parts through extruding successive layers of material. FDM is one of the most accessible and affordable types of additive manufacturing, though it creates parts with poor surface finish and has relatively low build speed.
Fused Deposition Modeling	FDM. An additive manufacturing process that builds parts through extruding successive layers of material. Fused deposition modeling is one of the most accessible and affordable types of additive manufacturing, though it creates parts with poor surface finish and has relatively low build speed.
Fused Filament Fabrication	FFF. An additive manufacturing (AM) process that builds parts through extruding successive layers of material, though it creates parts with poor surface finish and has relatively low build speed. Fused filament fabrication is one of the most affordable types of additive manufacturing and sometimes referred to as fused deposition modeling (FDM).

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

G Code	A programming language that uses letters and numbers to form commands for machine tools. G code programs are used to direct tool movements in additive manufacturing, CNC machining, and hybrid machining.
Hot-End Jam	A clog in the FDM extruder. Hot-end jams typically occur when the FDM extruder is not hot enough to melt the material correctly.
Industrial Internet Of Things	IIoT. A network of physical devices used in manufacturing that contain computing systems that allow them to send and receive data. The Industrial Internet of Things allows devices to exchange data and automate processes without human intervention.
Job File	A program created using build processor software that controls the functions of an FDM machine. Job file formats may be specific to each machine.
Lattice Structure	A repeating, symmetrical pattern of crossing strips of material that leave diamond- or square-shaped gaps between them. Lattice structures provide excellent strength to a part.
Lightweighting	In manufacturing, reducing the weight of a part or product without altering its function. Lightweighting results in parts and products that are less heavy than previous iterations.
Machine-To-Machine	M2M. The transfer of data between machines and the internet without human interaction. Machine-to-machine communication relies on sensors.
Magnification	A device held in front of the eye to enlarge an object for the viewer. Magnification can be used when conducting a visual inspection.
Material Canisters	A container that holds thermoplastic filament. Material canisters are used with FDM machines.
Micrometer	A U-shaped measuring instrument with a threaded spindle that slowly advances toward a small anvil. Micrometers are available in numerous types for measuring assorted dimensions and features.
Offset	A numerical value stored in the FDM control that repositions machine components. The OFFSET key allows operators to enter tool length geometry and part zero locations.
Personal Protective Equipment	PPE. Any clothing or device worn to minimize exposure to hazards or prevent injury. Common additive manufacturing (AM) personal protective equipment includes gloves, safety goggles, and nonflammable clothing.
Platen	The flat, built-in surface of the build chamber. Platens typically hold a build sheet or build tray.
Platen	The surface on which parts are built in an FDM machine. Platens typically hold a build sheet or build plate.
Polyphenylsulfone	PPSF. A thermoplastic polymer with strong mechanical performance and high resistance to heat and chemicals. Polyphenynylsulfone is extremely resistant to corrosion.
Pre-Processing	A group of procedures that are used to clean, improve, or otherwise prepare a part for use by a manufacturer or consumer. Pre-processing for additively manufactured parts includes preparing a build sheet for printing in fused deposition modeling.
Purge Bucket	On an FDM machine, a container that receives waste collected from the debris chute. Purge buckets should be emptied daily.
Purge Chute	A channel in the the back of the build chamber used to dispense waste. Purge chutes on FDM machines typically guide waste to the bottom of the build chamber or purge bucket.
Rapid Prototyping	A product development technique in which additive manufacturing (AM) methods are used to create prototypes for a traditional manufacturing operation. Rapid prototyping allows engineers to quickly create a number of prototypes in a short time period, reducing lead time.
Scraper	A tool with a handle and a wide, flat blade. Scrapers are used in additive manufacturing to separate plastic parts from build platforms.
Sensors	A device within a device that detects a physical stimulus and turns it into a signal that can be measured or recorded. Sensors in smart devices make it possible for machines to communicate digitally.
Setup	All the necessary preparation of tooling and fixturing that occurs on a machine before operation begins. Proper machine setup is an important factor in optimizing tool life.
Smart Manufacturing	An information-driven, event-driven, collaborative orchestration of business and factories across the value chain. Smart manufacturing uses sensors, data, and analytical models to automate and prescribe action for various processes and situations.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.

action for various processes and situations.

Vacuum A suction tool used to collect loose dust, dirt, and debris. A vacuum is also used to hold the build sheet in place on some FDM machines.

Z Axis The linear axis that represents the up-and-down motion of an object relative to the origin. The Z axis is the only vertical axis in the Cartesian coordinate system.

Copyright © 2023 Tooling U, LLC. All Rights Reserved.